

Computer Networking with Basics Technology to Advanced and Complete Knowledge For Academic level

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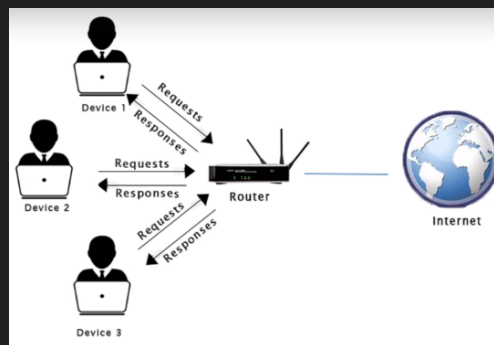
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Introduction

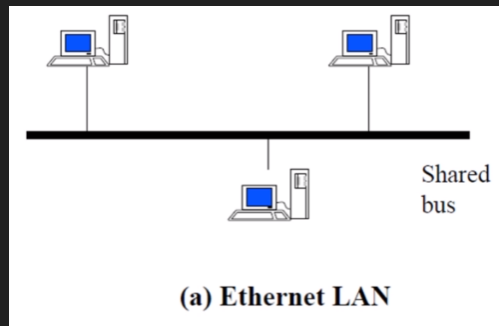
1.1 What is a network?

- A network can be defined as a group of computers and other devices connected in some way to be able to exchange data.

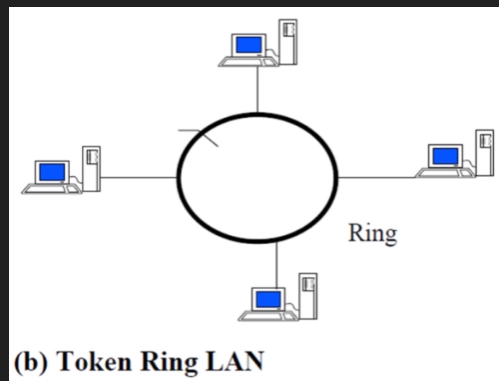


1.2 Types of networks and its advantages

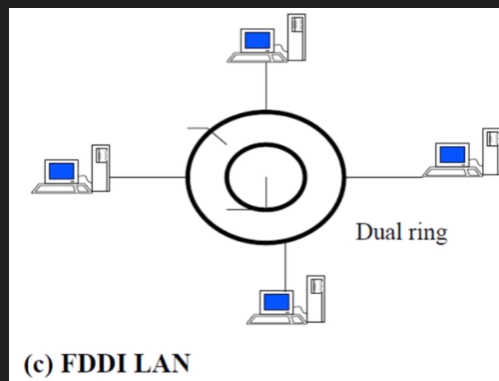
- There are 2 principle kinds of networks: wide area networks WAN and local area networks LAN.
- WAN:
 - Cover cities, countries, and continents.
 - Based on packet switching technology.
 - Examples of wan technology: asynchronous transfer mode (ATM), integrated services digital network (ISDN).
- LAN:
 - Cover buildings or a set of closely related buildings.
 - Examples of LAN technology: ethernet, token ring, and fibber distributed data interconnect (FDDI).
 - Ethernet LANs based on a bus topology and bradcast communication.



- Token ring LANs: based on ring topology.



- FDDI LANs: use optical fibers and an improved token ring mechanism based on two rings flowing in opposite directions.



- Advantages of networks:
 - It enhances communication and availability of information.
 - It allows for more convenient resource sharing.
 - It makes file sharing easier.
 - It is highly flexible.
 - It is an inexpensive system.
 - It increases cost efficiency.
 - It boosts storage capacity.

Chapter 2

Computer Network Basics

2.1 Computer networks

- A computer network is a system in which multiple computers are connected to each other to share information and resources.
- Characteristics of a computer network:
 - Share resources from one computer to another.
 - Create files and store them in one computer, access those files from the other computers connected over the network.
 - Connect a printer, scanner, or a fax machine to one computer within the network and let other computers of the network use the machines available over the network.
- Hardware required to setup the computer network:
 - Network cables: ethernet cables.



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- Distributors:



- Router:



- Network card:



- Universal Serial Bus: USB:

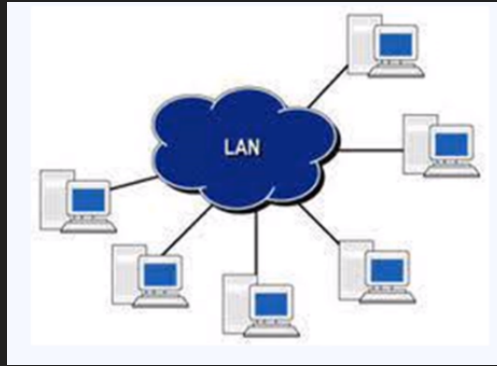


- Types of computer networks:

- There are basically 4 types: PAN, LAN, MAN, WAN.

- * PAN: Personal Area Network.
- * LAN: Local Area Network.
- * MAN: Metropolitan Area Network.
- * WAN: Wide Area Network.

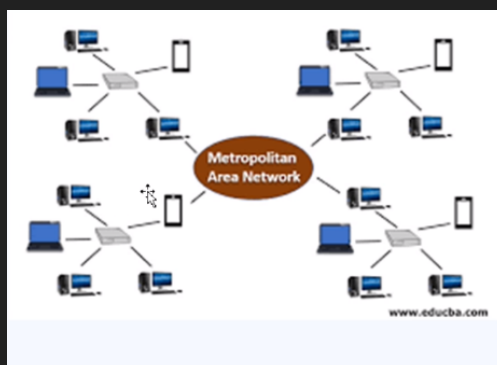
- LAN: stands for Local Area Network: it is a computer network that has 2 or more connected computers, and is contained within a small geographic area, usually within the same building. Home WiFi networks and small business networks are common examples.



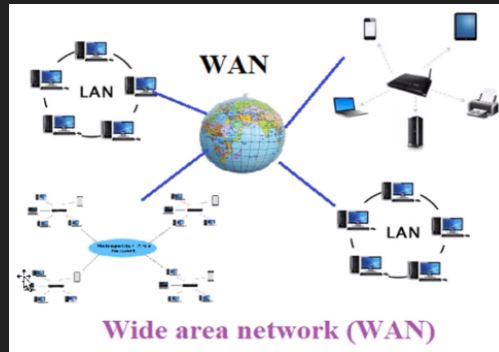
- PAN: Personal Area Network: is a computer network that connects computers and devices within the range of an individual person. A PAN provides a network range within a person's range typically within a range of 10 meters or 33 feet.



- MAN: A metropolitan area network is a network in which private data is exchanged, usually used by a single organization in several buildings that are all interconnected in the same geographic vicinity. It is larger than a LAN in a single building but not large enough to be considered a WAN. The size usually ranges from 5 kilometers to 50 kilometers. If all the buildings are in the same piece of contiguous property, it may also be considered a campus network.

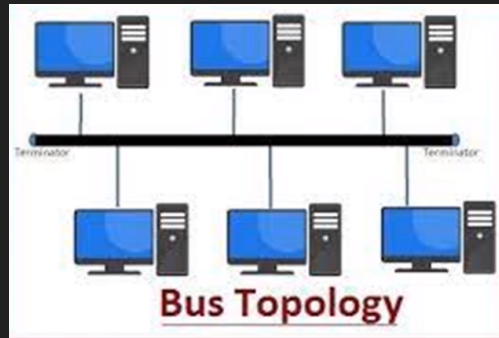


- WAN: Wide Area Network: is a geographical area consisting of 2 or more LANs or MANs. These networks are established with leased telecommunication circuits, in which two sides which are connected have routers that connect the LAN of both sides together in a network to facilitate communication.



2.2 Topology and it's types

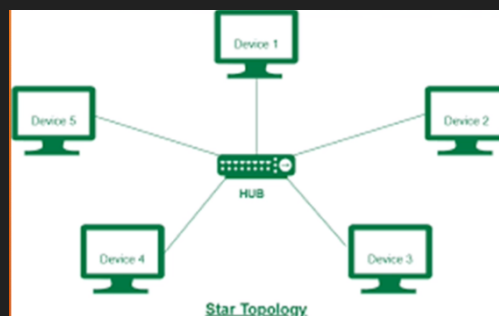
- A topology: is derived from two greek words, topo and logy. Where topo means place and logy means study. In computer networks, a topology is used to explain how a network is physically connected and the logical flow of information in the network.
- There are 2 types of topology.
 1. Physical topology.
 2. Logical topology.
- Now to explain the physical topology:
 - A physical topology describes the way in which the computers or nodes are connected with each other in a computer network. It is the arrangement of various elements (link, nodes, etc.) including the device location and code installation of a computer network. In other words... we can say that it is the phyical layout of nodes, workstations, and cables in the network.
- Logical topology:
 - A logical topology describes the way, data flow from one computer to anoter.
 - It is bound to a network protocol and defines how data is moved throughout the network and which path it takes.
 - In other words, it is the way in which the devices communicate internally.
- In a computer network, there are mainly six types of topology.
 1. Bus topology.
 2. Ring topology.
 3. Star topology.
 4. Mesh topology.
 5. Tree topology.
 6. Hybrid topology.
- Bus topology: A bus topology is the simplest kind of topology in which a common bus or channel is used for communication in the network. The bus is connected to various taps and droplines.



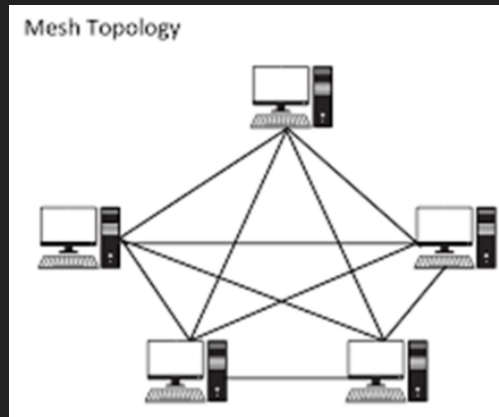
- It has certain advantages: for example it is simpler to set up and if one node fails it will not affect other nodes.
- Ring topology: in this one the computer is connected to exactly two other computers to form the ring. The message passing is unidirectional.



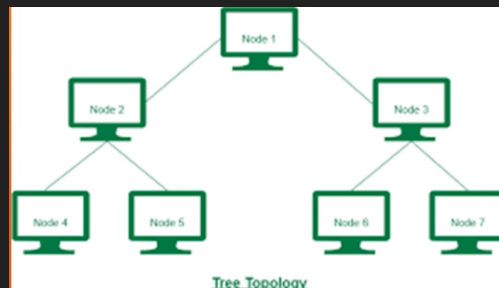
- The one disadvantage is that if one node fails the whole network fails.
- Star topology: is a computer network in which all the nodes are connected to a centralized hub.



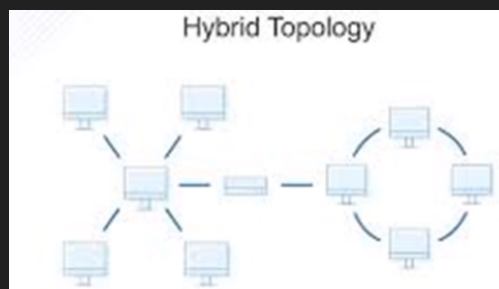
- This topology is better in terms of security.
- The data does not pass through every node.
- Easy to troubleshoot and easy to scale.
- Mesh topology: is a computer network topology in which nodes are interconnected with each other.



- In this topology there is a more direct line of communication. Much faster.
- If one node fails the network still works.
- Tree topology: is a computer network topology in which all the nodes are directly or indirectly connected to the main bus cable.



- It is a combination of a bus and star topology.
- Hybrid topology: topology in which there is a combination of two or more topologies.



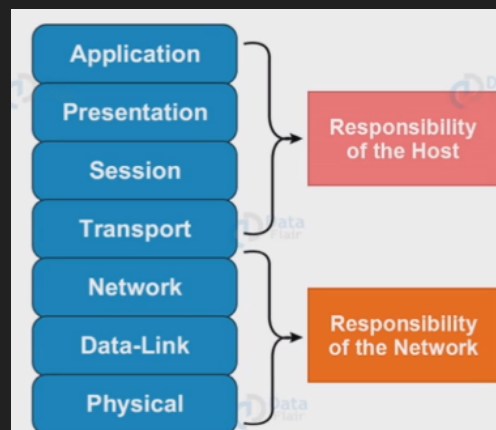
- These are the more used in the real world.
- Can handle a large number of nodes.
- Provides flexibility to modify the network according to our needs.

Chapter 3

Reference Models

3.1 OSI Model

- What is it?
 - OSI Model stands for Open System Intercommunication model.
 - OSI consists of seven layers, and each layer performs a particular network function.
 - OSI model was developed by the international organization for standardization (ISO) in 1984.
 - OSI model divides the whole task into seven smaller and manageable tasks.
 - Each layer is self contained, so that tasks assigned to each layer can be performed independently.
- Characteristics of OSI model:



- Divided in upper layers and lower layers. The responsibility of the host is the tasks assigned to upper layers, and the lower layers are the responsibility of the network.
- Functions of the OSI layers: There are the seven OSI layers. Each layer has different functions.
 - Physical layer: transmits raw bit stream over the physical medium.
 - Data-link layer: Defines the format of data on the network.
 - Network layer: Decides which physical path the data will take.
 - Transport layer: Transmits data using transmission protocols including TCP and UDP.
 - Session layer: Maintains connections and is responsible for controlling ports and sessions.
 - Presentation layer: Ensures that data is in a usable format and is where data encryption takes place.
 - Application layer: Human-computer interaction layer, where applications can access the network services.

3.2 Transmission Control Protocol (TCP/IP)

- What is it?
 - TCP is a connection oriented communications protocol that facilitates the exchange of messages between computing devices in a network. It is the most common protocol in networks that use the Internet Protocol (IP); together they are sometimes referred to as TCP/IP.
- TCP/IP model:
 - The TCP/IP model was developed prior to the OSI model.
 - The TCP/IP model is not exactly similar to the OSI model.
 - The TCP/IP model consists of five layers: the application layer, transport layer, network layer, data link layer and physical layer.
 - The first four layers provide physical standards, network interface, internetworking, and transport functions that correspond to the first four layers of the OSI model and these four layers are represented in TCP/IP model by a single layer called the application layer.
 - TCP/IP is a hierarchical protocol made up of interactive modules, and each of them provides specific functionality.
- Layers of the TCP/IP model:
 - The TCP/IP model is structured with four different layers. These four layers are:
 1. Network access layer.
 2. Internet layer.
 3. Host to host layer.
 4. Application layer.
- Network Access layer:
 - This is the bottom layer of the TCP/IP model architecture.
 - It is a combination of the data link and physical layer of the OSI model.
 - The physical transmission of data takes place at this layer.
 - Once the frames are transmitted by a network, encapsulating the IP datagram into these frames is done in this layer.
 - Also, the mapping of IP address into physical address is done here.
- Internet layer:
 - It is the second layer of the TCP/IP model and this layer is parallel to the network layer of the OSI model in terms of structure.
 - Sending the data packets to their destination network is the main function of the internet layer.
 - The logical transmission of data takes place at this level.
 - There are three different protocols used in this layer.
 - These include: IP (Internet Protocol), ARP (Address Resolution Protocol), ICMP (Internet Control Message Protocol).
- Host-to-host layer:
 - This layer is parallel to the transport layer of the OSI model.
 - The error-free delivery of data is the main function of this layer.
 - There are 2 main protocols present in this layer: